230 A Human-Like Action Prediction Method

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1 Introduction

Being society aging, an intelligent room is needed for the aged or handicapped. Such systems need to have high prediction accuracy, high dynamic adaptation capability and high flexibility. One of the most important problems is how to predict user actions because each user behaves differently. In this paper we describe how to solve these problems in an intelligent room that we called learning room.We have proposed a method to predict user action using Inductive Learning(IL)[1] with N-gram. The system based on our proposed method is able to acquire the immanent causality rules automatically from data pairs of continuous room states by means of IL.

A possible solution to this problem is to use analytical approach, furnish the system with rules in advance. However, since habits and preferences vary from one user to another, to prepare these rules requires huge labor for each user.

Another idea is to use statistical approaches that can be classified into Point-Based Prediction and Pass-Based Prediction. Point-Based draw on relatively small amount of information from data and therefor prediction can potentially be rather inaccurate. There has been some Path-based models prediction system. These models are built based on the user's previous path data using N-gram. The problem is that prediction becomes impossible when the predicted target appearance frequency is low, and it is difficult to realize dynamic adaptation.

In this paper, we describe our basic idea how to realize an intelligent learning room system. As unified with IL and N-gram, the system demonstrates a high predictive accuracy. However, the acquired rules express the user's habits and preferences, the system is possible to adapt dynamically to each user. The user needs to proofread the errors in prediction results. Therefore the prediction ability improves. As a result, the number of errors decreases. This paper unifies N-gram and Inductive Learning to develop a Point-Pass Based Prediction system. The system was found to have good accuracy of which the highest prediction accuracy was about 89.3(%).The system was improved that it has high dynamic adaptive ability.

2 Inductive Learning with N-gram

K.Araki et al. have proposed Inductive Learning(IL) as a natural language processing approach. IL is a Point-Point-Based method and the effectiveness had been proved . The basic idea of IL is reflexive extraction different parts and common parts from similar data pairs. System can be automatically acquire rules in multistage steps by means of this idea.

As it is well known, N-gram is a Pass-based method for machine learning. A N-gram model also being called language model, it is widely used in speech recognition or prediction system. We unify the IL and N-gram according to our difined dynamic hash table, as shown in Table 1.

3 Outline

To simplify analysis, we assume our system consists of eight artificial elements, four natural elements. Each element has some initial fixed values.

The user actions need to be accessible in some tangible form, and then we can generate a sequence of historic actions as a string of action symbols or we can represent individual actions as separate states with corresponding state transitions.

$$States_k^n = (N_k^n, A_k^n)^1 \quad (1)$$

(N_k^n) expresses natural elements vector space, and (A_k^n) means artificial element vector space as follows:

$$N_k^n = (Week_k^n, Time_k^n, Brightness_k^n, Temperature_k^n) \quad (2)$$

$$A_k^n = (Dk^n, TV_k^n, P_k^n, C_k^n, L_k^n, AC_k^n, Tel_k^n, Al_k^n)^2 \quad (3)$$

According to the basic idea of Inductive Learning,by extracting the different part from artificial elements states, we can acquire initial rules as follows.

$$Rule_k^n = (S_k^n, Action_k^n) \quad (4)$$

In order to acquire rule adaptation capability, we do abstraction process for each initial rule using IL. For N-gram processing, we extract user action sequence as scripts. In this paper, we use trigram to evaluate our peoposed method.

Figure 1 shows the procedure of our proposed method, it consists of prediction process, proofreading process, feedback process and learning process.

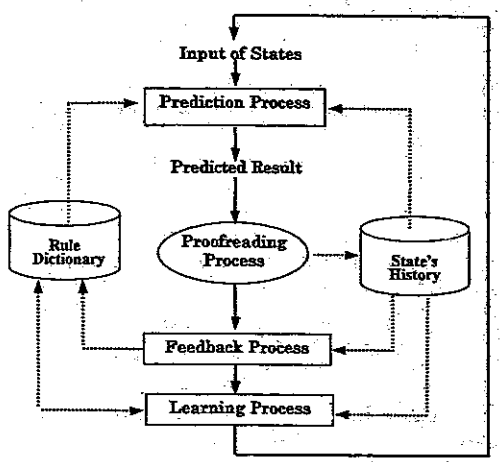


Figure 1 Outline of the Procedure

In prediction process, system makes prediction by pattern matching, while referring to the data history

[1] "n" is the number of data order, "k" express date of data.

[2] Express:door,TV,program,curtain,light,airconditioner,telphone, alarm clock respectively.

Table 1　Dynamic Table for Prediction

$...rule^{n-2}rule^{n-1}rule^{n}$	IL credibility of $rule^{n}$	N-gram information	unified information of IL and N-gram

and the rule dictionary. In proofreading process, the user judges whether the action is correct or not. In feedback process, rules credibility is updated based on four values tracked each rules those values are correct fitness frequency (CF), error fitness frequency (EF), correct fitness rate (CR) and error fitness rate (ER) of rule. The credibility evaluation function (CEF) of our proposed method is defined as follows:

$$CEF = CR - \alpha * ER + \xi * \beta * P_n \tag{5}$$

$$\xi = \begin{cases} 1 & (\text{Frequency} \geq \text{Threshold}) \\ 0 & (\text{Frequency} < \text{Threshold}) \end{cases} \tag{6}$$

,where: α and β are coefficients.
CR and ER are defined as follows:

$$CR = \frac{CF}{CF+EF} \quad ER = \frac{EF}{CF+EF} \tag{7}$$

The learning process consists of the following steps: acquisition of initial rules, rule abstraction process, acquisition of action scripts, analysis of probability with trigram, and rule information updating. This implementation provides the possibility that the system is able to adapt dynamically to each user. When performing rule abstraction process, We gave some range for each natural element to decide the criteria of common parts.

For rule abstraction process, we defined abstraction degrees for artificial and nature element, which are shown as follows:

$$Dan = \frac{X}{N} \quad Daa = \frac{Y}{M} \tag{8}$$

,where: Dan is the degree of abstraction for the natural elements in the rules. X is the number of variables in the natural elements portion. N is the number of natural elements. Daa is the degree of abstraction for the artificial elements in the rules. Y is the number of variables in the artificial elements portion.M is the number of artificial elements.

4　Evaluation Experiment

The system based on our method was developed for experiment to investigate the validity. Since our aim is to propose an adaptive approach, we will evaluate online performance. We do not perform cross-validation evaluation.

The data was collected based on the person actual life, from five graduate students of graduate school of engineering. We generated about 16,000 data that corresponded to a period of 91 days. In order to keep the starting state constant for each user, the data history and the rule dictionary always started from empty initial state.

We define precision as follows:

$$Precision = \frac{P^{+}}{P^{+}+P^{-}} * 100(\%) \tag{9}$$

,where: P^{+} is the number of correct prediction results. P^{-} is the number of incorrect prediction results.

In order to optimize the coefficients of the credibility evaluation function, we did some preliminary experiments. We applied the Greedy Method [2] to all collected data of C to obtain the optimal coefficients for the credibility evaluation functions.In the formula (5) the parameters were found to be 10 and 50 respectively. For the abstracted rules,Both the value of Dan and Daa are 0.5 by preliminary experiments.

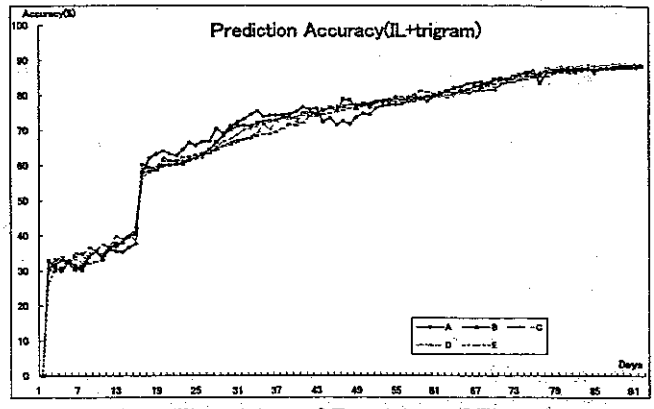


Figure 2　Transition of Precision of IL+trigram

5　Results and Discussion

Figure 2 shows the performance of our proposed method.We just use the rank1 of prediction result to evaluate the accuracy of our experiment. We found that the highest prediction accuracies for users are about 89.1% (A), 88.6% (B), 89.3% (C), 88.3% (D), and 88.7% (E). It is shown that our method had good prediction performance and good dynamic adaptive ability.

With these results, We can draw two conclusions from the experiment results:

(1) Our method does not need large database in advance.

(2) The Point-Pass-based prediction method validity was proved.

6　Conclusions and Future Work

In this paper we have described our approach to predicting a user actions in an intelligent system such as a learning room.It is showed that our system performance has good accuracy.Furthermore, we also observed that our system could adapt dynamically to any user, since the data history and the rule dictionary were always initially empty.

In our future works, we hope to certify the validity of out method using above 4-gram, to prove the flexibility of the system with multiuser, to try using the HMM Model to increase the prediction accuracy.

Reference

[1] K. Araki and K. Tochinai.“Effectiveness of Natural Language Processing Method Using Inductive Learning”,Proceedings of the IASTED International Conference ARTIFICIAL INTELLIGENCE AND SOFT COMPUTING, May, 2001, Cancun, Mexico, pp.295-300.

[2] T.Asano, H. Imai, Computation and Algorithms, Ohm Publish 2000, In Japanese.